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How much do zooplankton and fine-scale processes matter for the ocean biological pump?

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The ocean biological carbon pump stores carbon away from the atmosphere through multiple pathways, including gravitational settling, physical transport, and organism vertical migration. Robust assessments of its magnitude remain challenging. Traditional approaches often quantify pathways separately, risking double-counting, missing high-frequency processes such as fine-scale physics and daily vertical migration over large scales, and emphasizing particulate carbon while overlooking non-particulate fluxes. Here, we apply a unified framework that quantifies all pathways simultaneously in a high-resolution (3 km) idealized North Atlantic model resolving seasonal biophysical dynamics, including zooplankton migration, from kilometer-scale fronts to regional biomes. We show that fine-scale physical and migrant pumps together contribute 15-20% of carbon storage. Their storage is dominated by non-particulate carbon fluxes (dissolved organic carbon transport, zooplankton respiration). Fine-scale dynamics modulate export depth and storage timescales both directly, through physical transport, and indirectly, by shaping fast-sinking carbon-rich filaments and controlling zooplankton migration depth. Adequately measuring and representing these pathways in next-generation observations and models is key to quantify carbon storage and its response to variability.

The ocean's biological carbon pump is a critical engine of the climate system, storing carbon away from the atmosphere[1,2]. This carbon storage occurs through a set of interconnected processes—referred to as 'pumps'—exporting and remineralizing both particulate and dissolved organic carbon in the ocean interior (Fig. 1a,S1)[3]. Current estimates suggest these pumps export between 5 and 12 PgC yr^{-1} from the surface ocean[4–8] and store between 1100 and 1500 PgC in the ocean interior[8,9]. However, none of these estimates fully capture all the pumps and their spatial and temporal scales, hindering our ability to accurately assess the biological carbon pump's current magnitude, its response to climate change, and its role as a climate feedback – reinforcing or dampening of climate change by biological pump changes[10].

A comprehensive assessment of the biological pump requires disentangling intricate processes that export both particulate and dissolved organic carbon on different time and space scales. The most well-studied pump is the gravitational pump, which is fueled by sinking particles (Fig. 1b). The physical pumps transport both particulate and dissolved carbon through three mechanisms (Fig. 1c): the large-scale pump transfers carbon via Ekman transport and overturning circulation[11], the mixed-layer pump exports carbon through seasonal mixed layer deepening and restratification[12,13], and the eddy subduction pump through turbulent ocean dynamics, including fine-scale eddies and fronts (< 200 km)[14,15], a pathway that has only been recently studied. Fine-scale dynamics also modulate upper-ocean stratification,

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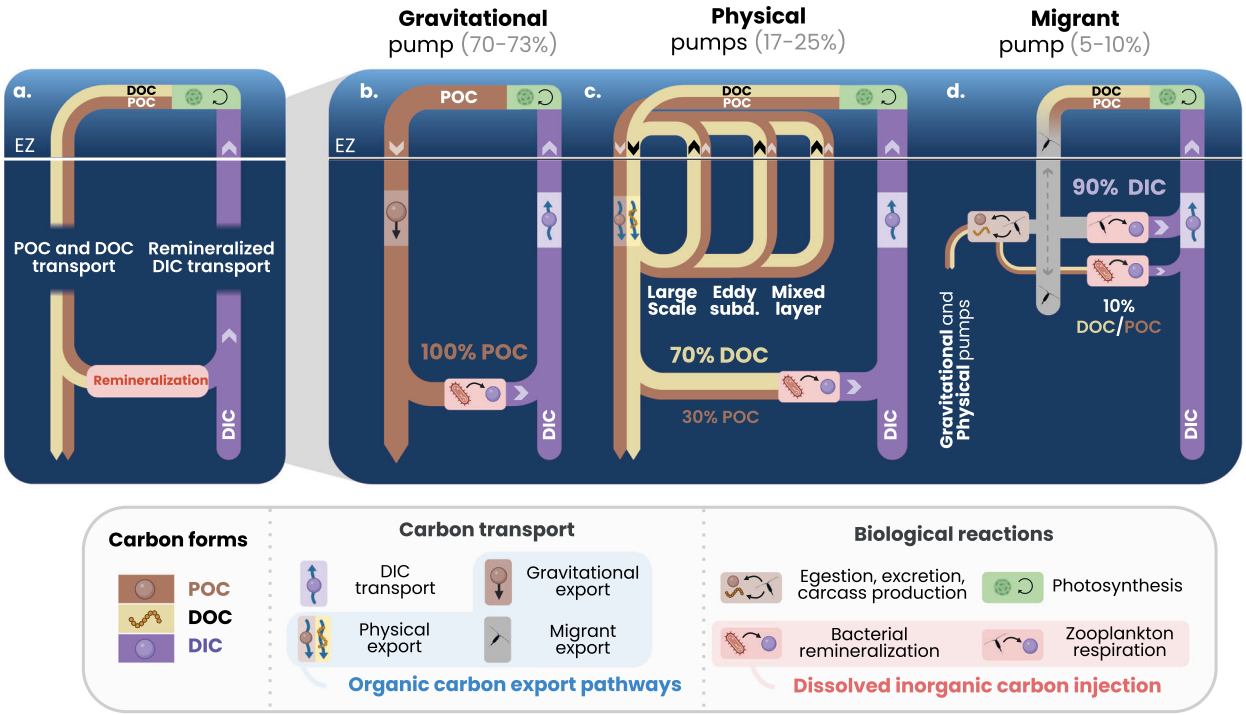


Fig. 1 | Unifying framework of the biological carbon pump. Biological carbon pump as represented in COBALTv2-DVM. **a** The biological carbon pump encompasses the (**b**) gravitational, **c** physical, and **d** migrant pumps. The physical pumps include the large-scale pump, the eddy-subduction pump, and the mixed-layer pump. These pumps act to increase the vertical DIC gradient via the production, export, and remineralization of organic carbon, while the upward transport of remineralized DIC acts to decrease this gradient. Organic carbon is produced in dissolved and particulate form in the euphotic zone (EZ) through photosynthesis (green block), and is exported by the pumps. POC (brown) is transferred by gravitational settling, physical export, egestion of fecal pellets, and the carcasses of migrating zooplankton. DOC (yellow) is transferred by the physical export and the excretion of migrating zooplankton. Remineralization of POC and DOC by bacteria and respiration by migrating zooplankton (red blocks) inject DIC (purple), which is stored in the interior ocean before being upwelled to the surface, closing the loop. Note that the arrow thickness is not conservative and proportional to the flux magnitudes. The schematic focuses on the component of the biological pump represented in the COBALTv2-DVM model; other pathways (e.g., dark CO_2 fixation via chemoautotrophy[77,78]) are not included. This figure was created using BioRender.com (https://BioRender.com/kx5xjz5).

nutrient supply, primary production, and trophic interactions[15–18], thereby shaping the biological carbon pump beyond carbon transport alone. Finally, the migrant pump is powered by vertically migrating organisms, such as zooplankton, which actively transport particulate and dissolved carbon through egestion, excretion, and respiration along their migration path (Fig. 1d)[19–21]. While zooplankton vertical migration has been described for 200 years[22], its contribution to the biological carbon pump remains incompletely constrained despite recent advances. Full assessments are challenging due to the vast range of scales that these pumps span. Zooplankton migration occurs daily over hours and hundreds of meters[23], climate-driven circulation changes extend over decades and global scales[7,24,25], and in between, submesoscale fronts[14,17] and mesoscale eddies[15,26,27] act on spatial scales of 1-100 km and temporal scales from days to months. This complexity and variability pose significant challenges for existing observational and modeling tools[28].

Current evaluations remain limited by the partial characterization of the pumps and the poor resolution of seasonal and fine-scale dynamics, leading to fragmented and incomplete estimates. Observational approaches often overlook non-particulate carbon fluxes, which are harder to measure than particulate fluxes[29–31] and tend to largely exclude suspended and ascending particles[32]. Model studies either represent zooplankton migration using simplified parameterizations[8,20,33] that fail to capture its full complexity, or omit it due to computational and conceptual trade-offs[7,15,24]. Typically, studies focus on a single pump at a time[29,34–36], without isolating the contributions of other pumps, potentially causing double counting when aggregating individual estimates[3,37]; e.g., migrant-produced fecal pellets can be counted in both the migrant and gravitational pumps. Estimates are also constrained in spatio-temporal scales. Observations provide valuable snapshots in time and space but are challenging to extrapolate at regional and seasonal scales. Meanwhile, many global and regional models fail to resolve fine-scale processes such as eddies and fronts, playing a critical role in carbon export[15,38]. Finally, while most studies focus on export fluxes, they rarely consider the depth at which this carbon is remineralized by bacterial breakdown or zooplankton respiration, which is key in determining the timescale over which it remains stored in the interior ocean[2].

Here, we address these challenges by quantifying how much organic carbon leaves the sunlit surface ocean (export, gC m^{-2} yr^{-1}) and how much is stored in the ocean interior as dissolved inorganic carbon (storage, gC m^{-2}) by the different components of the biological carbon pump, using an idealized high-resolution model of the North Atlantic Ocean (DG-MOM6-COBALTv2-DVM[39]). This region covers contrasting biomes featuring marked seasonality and fine-scale dynamics[15,38], and supports active gravitational, physical, and migrant carbon pumps[13,19,40,41], offering a representative setting to understand their contribution to global carbon export and storage. Our model explicitly resolves these three pumps and covers their spatio-temporal dynamics from hourly and kilometric scales to seasonal and regional scales. We show that the physical and migrant pumps store more carbon via the export of non-particulate carbon (dissolved organic carbon transport, migrating zooplankton respiration) than via particle export, highlighting the need to better account for these fluxes in biological carbon pump assessments. Our results further show that fine-scale processes shape carbon export

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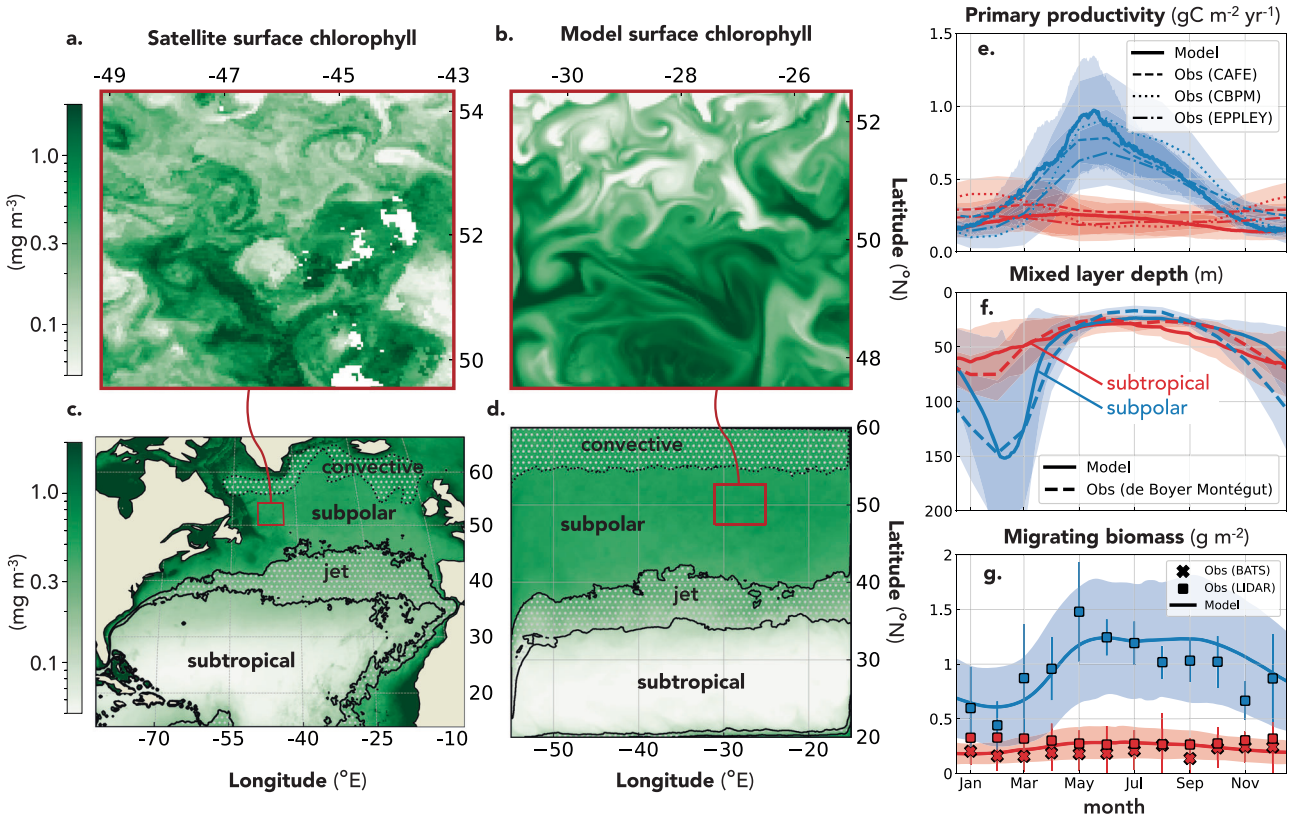


Fig. 2 | Observed and simulated North Atlantic biophysical dynamics. Subpolar surface chlorophyll concentration snapshot from (**a**) satellite observations and **b** the double gyre model on March 10th. Mean annual surface chlorophyll concentration from (**c**) satellite observations and **d** the double gyre model. Seasonal cycle of (**e**) net primary productivity, **f** mixed layer depth, and (**g**) migrating zooplankton biomass from observations (dotted lines) and in the model (solid lines), for the subpolar (blue) and subtropical (red) biomes. Shadings show the spatial variability within each biome (two standard deviations). Subtropical and subpolar biomes are defined by annual chlorophyll concentrations lower than 0.15 mg m^{-3} and higher than 0.35 mg m^{-3}, respectively (black lines in a, b). Deep convection region defined by an annual mixed layer depth >150 m, and the jet region defined by chlorophyll concentrations between 0.15–0.35 mg m^{-3}, are excluded from the biome analysis (dotted areas in **a**, **b**). Sources of the observations are detailed in the method section.

and storage directly via the eddy-subduction pump, but also by modulating the export magnitude and depth of the other pumps. We argue that adequately characterizing biophysical complexity –including fine-scale processes, biological migration, and non-particulate carbon fluxes– is key to capturing current ocean carbon storage and its response to future changes.

Results and Discussion

Carbon flux contrasts in the North Atlantic

Our double gyre model reproduces the observed fine-scale and large-scale biophysical dynamics that structure the North Atlantic biological carbon pump, including primary productivity, migrating zooplankton biomass and vertical winter mixing (Fig. 2, see details in Poupon et al. 2025[39]). Specifically, the model captures the variability associated with turbulent dynamics, as shown by kilometric-scale chlorophyll filaments, reflecting fine-scale variations in primary productivity, and by eddy kinetic energy patterns, both matching satellite observations (Fig. 2a, b,S2). The model also simulates the regional contrast between the high-productivity, high-seasonality subpolar biome (annual surface chlorophyll >0.35 mg m^{-3}) and low-productivity, low-seasonality subtropical biome (annual surface chlorophyll <0.15 mg m^{-3}, Fig 2c, d). In the subpolar biome, primary productivity peaks between May and June (0.92 gC m^{-2} d^{-1} in model and 0.65–0.93 gC m^{-2} d^{-1} in satellite, Fig. 2e), followed by migrating zooplankton biomass that peaks between May and September (1.2 g m^{-2} in model and 1.4 g m^{-2} in observations, Fig. 2g), with both reaching their minimum in winter (0.14 gC m^{-2} d^{-1} and 0.7 g m^{-2} in model, 0.10–0.24 gC m^{-2} and d^{-1} and 0.4 g m^{-2} in satellite, Fig. 2e, g). In contrast, in the subtropical biome, primary productivity and zooplankton migrant biomass remain relatively low and stable year-round (0.17–0.25 gC m^{-2} d^{-1} and 0.1–0.3 g m^{-2}, in both model and observations, Fig. 2e, g). The model also captures the contrasting physical dynamics between the subpolar biome, where the mixed layer depth strongly varies from shallow summer (25–35 m) to deep winter (120–140 m) values, and the subtropical biome, where it stays relatively shallow (30–60 m, Fig. 2f).

These regional biophysical differences drive high carbon export and storage contrasts between subpolar and subtropical biomes. The simulated annual carbon export (defined as net carbon exported and remineralized below the euphotic zone, see Methods) in the high-productivity subpolar biome is 3 times greater than in the low-productivity subtropical biome (36 vs 12 gC m^{-2} yr^{-1}, Fig. 3a,S3). This contrast is further amplified because, for any given depth at which remineralization occurs, carbon remains stored for longer in the subpolar biome than in the subtropical biome (based on storage times from the Ocean Circulation Inverse Model[8], see Fig. S4 and Methods). As a result, carbon storage is 3.6 times greater in the subpolar biome than in the subtropical biome (5.8 vs 1.6 kgC m^{-2}, Fig. 3c). Despite these differences, we show in the following that relative contributions of the different pumps to export and storage, and the contribution of non-particulate components are remarkably similar across the two biomes.

Carbon pump contributions to export and storage

Our analysis reveals that the relative contributions of the gravitational, physical, and migrant pumps to carbon export and storage are similar

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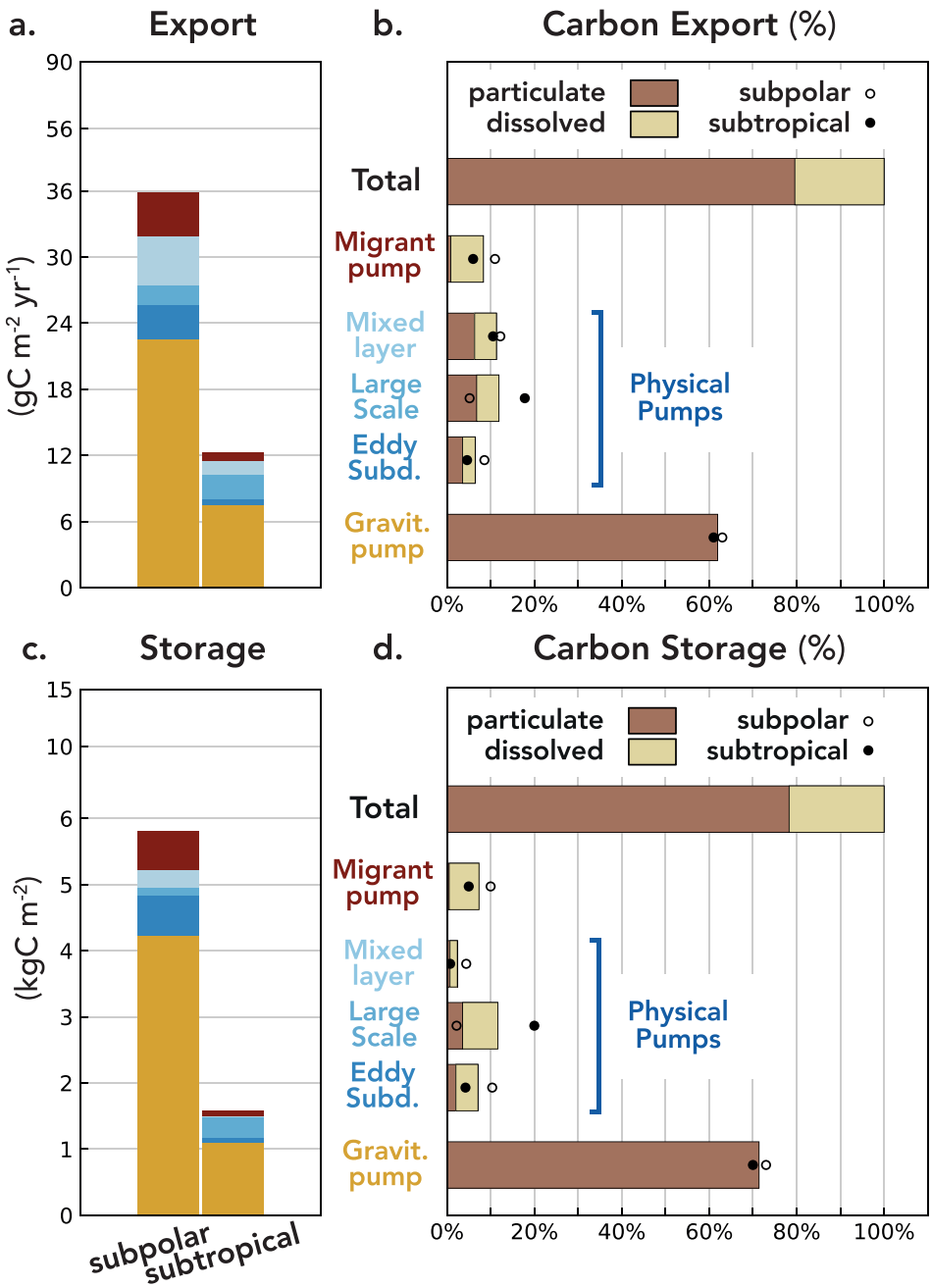


Fig. 3 | Regional contrasts in particulate and non-particulate carbon export and storage. a Annual carbon export and **c** carbon storage in the subpolar (left bars) and subtropical (right bars) biomes, partitioned among the gravitational (yellow), physical (blues), and migrant (red) pumps. The physical pumps include the mixed layer (light blue), large-scale (blue), and eddy-subduction (dark blue) pumps. Relative contributions of each pump to (**b**) carbon export and **d** storage across the model domain (excluding convective region; see Methods). Pale yellow bars represent dissolved carbon fluxes, and brown bars represent particulate fluxes. Empty circles are the averages over the subpolar biome, and filled circles are over the subtropical biome.

across the subpolar and subtropical biomes (Fig. 3b). The gravitational pump dominates export (~60% in both biomes), followed by the physical pumps (~30%; mixed-layer: 10–12%, large-scale: 5–18%, eddy-subduction: 5–9%), and the migrant pump (~10%; subtropical: 6%, subpolar: 11%). Pumps contributions to storage differ from export (Fig. 3b, d). The gravitational pump is highly efficient, with a contribution to storage that exceeds that of export (~70% vs. ~60%). In contrast, physical pumps are less efficient and contribute less to storage than to export (ranging from 17 to 25% across biomes; mixed-layer: 1 to 4%, large-scale: 2 to 20%, eddy-subduction: 5 to 10%). The migrant pump contributes similarly in both biomes (subtropical: 5%, subpolar: 10%, Fig. 3b, d).

These differences in pump efficiency are controlled by the depth at which organic carbon exported by each pump is remineralized in our model (Fig. 3a–d). The deeper the remineralization –via bacterial breakdown of POC and DOC or zooplankton respiration (see Methods)– the longer the carbon remains stored, and the more efficient the pump (Fig. S4). Overall, most of the carbon stored by the biological pumps in the model is remineralized in the mesopelagic zone in both biomes (75% in the upper 1000 meters, 50% in upper 500 meters,

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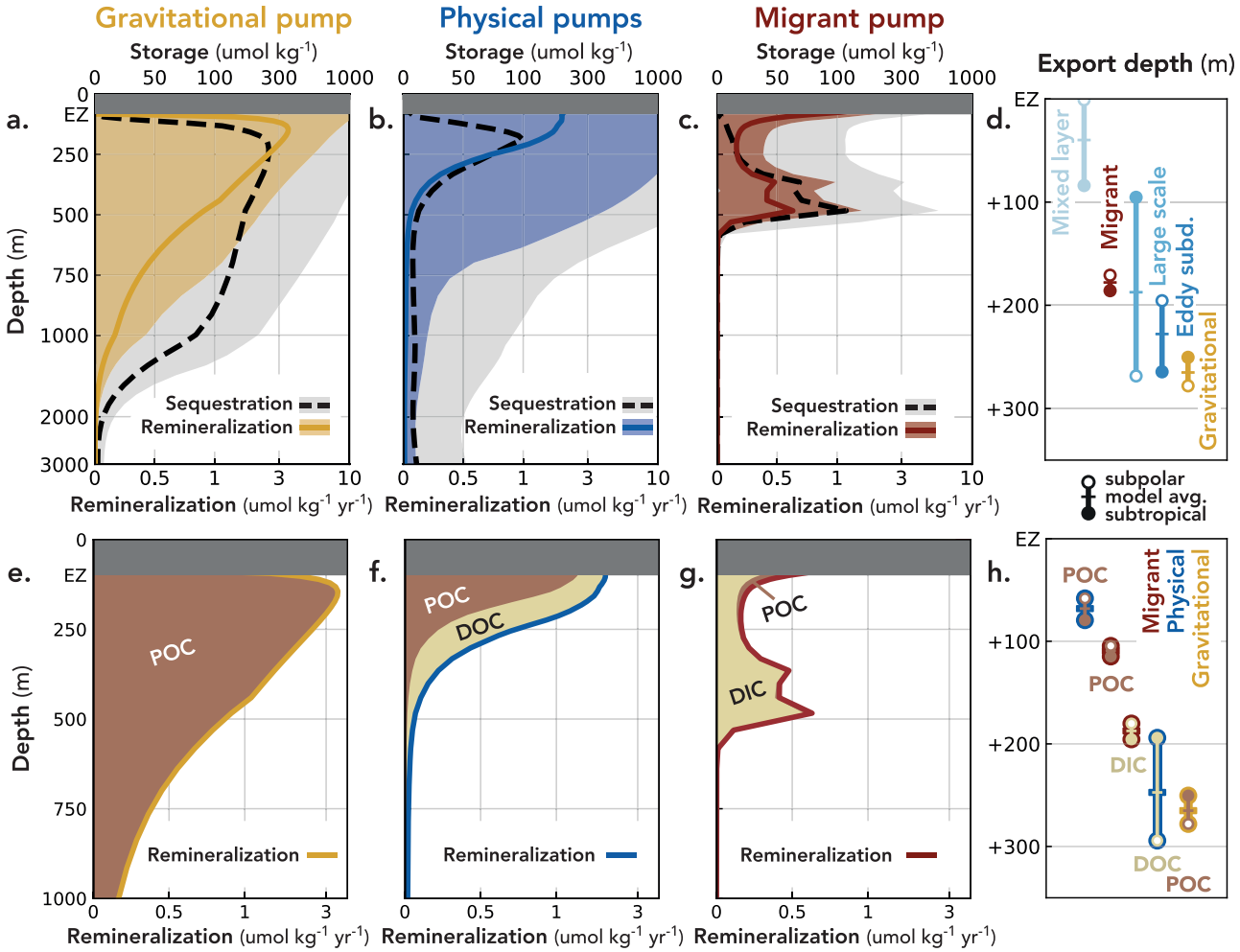


Fig. 4 | Depth profiles of carbon remineralization and storage. Annual-mean vertical profiles of carbon remineralization rates (solid yellow, blue and red lines) and storage (black dashed lines) associated with (**a**) gravitational, **b** physical, and **c** migrant pumps. Shading indicates ± 1 standard deviation of the annual spatial variability. The annual-mean euphotic zone (EZ, defined as the depth of 0.1% of surface irradiance) is shown in gray. **d** Export depth of carbon below the euphotic zone for the migrant pump (red), mixed-layer pump (light blue), large-scale pump (blue), eddy-subduction pump (dark blue), and gravitational pump (yellow). Horizontal lines show the model domain average (excluding convective region; see Methods); empty circles indicate the subpolar average and filled circles the subtropical average. **e**–**g** Annual-mean vertical profiles of carbon remineralization partitioned into particulate organic carbon (POC, brown) and dissolved carbon (pale yellow). **e** The gravitational pump (yellow line) transports only POC. **f** Physical pumps (blue line) transport both particulate and dissolved organic carbon. **g** The migrant pump (red line) injects dissolved inorganic carbon (DIC) through zooplankton respiration and particulate organic carbon (POC) through egestion and mortality; DOC excretion also occurs but is too small to be visible. **h** Export depth of particulate (brown) and dissolved (pale yellow) carbon below the EZ for the migrant (red), physical (blue), and gravitational (yellow) pumps.

Fig. 4a–d), consistent with recent findings[42]. However, the organic carbon exported by each pump is remineralized over distinct depth ranges. In both biomes, carbon exported by the gravitational pump is remineralized on average 250–280 m below the euphotic zone, making this pump the most efficient (Fig. 4a, d). Carbon exported by physical pumps is largely remineralized at shallower depths (Fig. 4b), similar to previous work[31], our model shows large differences between export pathways. The eddy-subduction pump exports carbon on average 200–270 m below the euphotic zone, while the large-scale pump exports it on average 100–270 m below and the mixed-layer pump only 0-80 m below (Fig. 4d). Finally, the carbon exported by the migrant pump is mainly remineralized 170-190 m below the euphotic zone (Fig. 4c, d). Thus, the eddy subduction pump and the migrant pump are highly efficient storage pathways that complement the gravitational pump.

Importance of non-particulate carbon fluxes

Fluxes of non-particulate carbon from the physical pump (DOC transport) and migrant pump (injection of DIC by zooplankton respiration) support more than 20% of the total simulated carbon export and storage in the subpolar and subtropical biomes (Fig. 3c, d). The physical transport of DOC contributes ~62% to the physical pump storage (eddy-subduction pump: 19%, large-scale pump: 39%, mixed layer pump: 4%), while that of POC, including plankton and sinking detritus, only contributes ~38% (Figs. 3d, S5). This large contribution of DOC to the physical pump storage is controlled by the different fates of exported DOC and POC in the water column. In fact, the physical pumps export more POC than DOC from the euphotic zone (14–19% vs 12–14% of total export, Figs. 3f and S5), but POC is more rapidly remineralized (~hours to weeks) than DOC (labile DOC: days, semi-labile DOC: months, semi-refractory DOC: decades), allowing DOC to be transported deeper by ocean currents before decomposition (+200–300m vs +60-80m below the euphotic zone, Fig. 4f, h). Specifically, remineralization of semi-refractory DOC accounts for most of the storage by the physical pump in the model (Fig. S5), making it a key pathway of the biological carbon pump.

Beyond the traditional emphasis on particulate fluxes[3], our results indicate that the migrant pump stores carbon primarily through the

injection of DIC via migrating zooplankton respiration. This respiratory flux accounts for 90% of the migrant pump carbon storage in our model (Figs. 3d, 4g, h). In contrast, POC egested by migrating zooplankton is largely confined to the euphotic zone (upper 100 m, Fig. S6). This is mainly because digestion and egestion of POC by zooplankton occurs rapidly –within 15 minutes to 3 hours after feeding (Fig. S7,[43]) while zooplankton are still in the upper ocean. In contrast, zooplankton continuously respire[44,45], injecting a larger amount of DIC at their migration depth (~300–500 m, Fig. 4g, h) where they spent 9–17 hours during daytime (Fig. S7), leading to longer storage times for the injected DIC than for the egested POC (Fig. 4g, h). This dominance is consistent with existing work[46]. We conclude that carbon storage by the non-particulate components of the physical and migrant pumps exceeds that of their particulate counterparts. This result further suggests that dissolved pathways warrant explicit consideration alongside particulate fluxes in assessments of these pumps[3,29,30].

Fine-scale dynamics shape carbon export and remineralization depths

Fine-scale eddies and fronts modulate both the magnitude and depth of carbon export, as illustrated by these model snapshots from March 5 (Fig. 5). Physical export shows the strongest fine-scale variability (Fig. 5b): within just a few tens of kilometers, intense export filaments (0.1–3 gC m^{-2} d^{-1}) are offset by equally strong re-entrainment into the euphotic zone (-0.1 to -3 gC m^{-2} d^{-1}), limiting the net export. Local estimates therefore, far exceed the regional net export, which cannot be meaningfully inferred without fully resolving these compensating fluxes. Gravitational export varies from 0.05 to 0.3 gC m^{-2} d^{-1} at fine scale (Fig. 5a), reflecting transitions between highly productive filaments—producing sinking particles such as aggregates and fecal pellets—and weakly productive ones (Fig. 2b). Migrant export also exhibits fine-scale variations on the order of 0.003–0.01 gC m^{-2} d^{-1}, driven by the heterogeneous spatial distribution of migrating zooplankton and their prey (Fig. S8). These patterns occur in both biomes and persist throughout the year (Fig. S9).

Our analysis shows that fine-scale dynamics also modulate the carbon export depth of each pump (Fig. 5d–f) and the associated carbon storage (Fig. 5g–i). Physical export depth is the most variable, ranging between +20 m and +450 m below the base of the euphotic layer over a few tens of kilometers (Fig. 5e), yet we do not detect a systematic small-scale covariation between export depth and export magnitude for the physical pump (Fig. 5h). In contrast, fine-scale dynamics modulate migrant export depth, from +50 m to +350 m below the euphotic zone (Fig. 5f), primarily by redistributing chlorophyll (Fig. 2b). In highly productive filaments, chlorophyll-induced shading locally reduces light levels, shoaling zooplankton migration and the depth at which DIC is injected by respiration[39]. As a result, hotspots of migrant export tend to coincide with shallower export depths (Fig. 5i), which limits storage. Gravitational export depth also varies at fine scales, from about +150 m to +400 m below the euphotic layer (Fig. 5d), but in this case hotspots of gravitational export tend to coincide with deeper export depths (Fig. 5g), which could reflect more effective particle ballasting and shielding in these filaments (Fig. S8) that allow aggregates to sink farther before remineralization and thus enhance storage. Annually, these fine-scale covariations reduce migrant pump storage by about ~5% in both biomes, while increasing gravitational pump storage by about 0.5–1.5% (Fig. S12). Together, these patterns show that fine-scale dynamics do not merely transport and redistribute carbon, but selectively amplify or reduce carbon storage by the different components of the biological carbon pump.

Bringing biophysical complexity into future observational and modeling paradigms

In both subtropical and subpolar biomes, we show that carbon storage by the migrant and physical pumps is dominated by non-particulate carbon fluxes, which together account for about 20% of total storage. In practice, these so-called "particle injection pumps"[3] are in fact mainly non-particulate pumps: the migrant pump via respiration and the physical pumps via the transport of DOC, suggesting a refinement in how these pathways should be considered. Our results are consistent with previous work[8], who showed that DOC dominates the large-scale pump yearly. Here, we show that this is also the case at fine scales, where the eddy-subduction pump stores most of the carbon via DOC transport. We also confirm large offsets between upward and downward fluxes previously reported[15]. This matters because current observational studies on the eddy-subduction export mostly focus on POC and do not take these limitations into account[29,37]. Current approaches have been developed to estimate DOC transport from floats[47,48], as well as upward/downward sub-mesoscale transport[49], and should be leveraged to evaluate their global impact. Additionally, recent progress in the understanding of microbial processes that remineralize organic carbon[50,51] and alter its reactivity[52,53] paves the way to better constrain its fate in the interior ocean. Our estimates of migrant export are lower than some previous work[21], which also consider higher trophic level migration. However, the ecological and biogeochemical models were not coupled—an approach that tends to overestimate fluxes, as past studies have shown[54,55]. Estimates based on simple parameterization embedded in biogeochemical models tend to lie much closer to our results[20,33]. A refinement of these estimates of migrating zooplankton respiration should account for coupled and extended size spectra, incorporating allometric respiration rates and migration depths[56], as well as physiological cyclicity associated with internal circadian rhythms[57].

We show that both the eddy-subduction pump and the migrant pump sustain deep carbon export and remineralization, making them efficient storage pathways complementing the gravitational pump. For instance, both of them account for about ~ 10% of carbon storage in the subpolar biome. We further demonstrate that fine-scale dynamics regulate carbon storage, not only via the eddy-subduction pump, but also by shaping the spatial patterns and covariance in export magnitude and export depth for both the gravitational and migrant pumps, modulating their respective storage. Current Earth system models—routinely used to assess the future evolution of the biological carbon pump[24]—do not represent these mechanisms[58]. They are primarily tuned to reproduce the bulk global export flux below the euphotic zone, which they may match for the wrong reasons, while misrepresenting the remineralization of this flux, the efficiency of the different pumps, and deep-ocean carbon storage[59,60]. For example, if zooplankton do not migrate, respired carbon remains in the surface layers, and carbon that could have been exported to depth by fine-scale processes is instead transported only shallowly by the mixed-layer pump, reducing carbon storage[11,56]. In addition, these export pathways are driven by distinct mechanisms whose responses to future climate change can be opposite in sign and remain highly uncertain[10]. Given the demonstrated importance of these pathways and their interactions, we argue that they must be incorporated into Earth system models[61], most likely through parameterizations informed by multiscale modeling and designed to respond consistently to future changes in stratification, productivity, and temperature. Without this, projections of future ocean carbon storage will remain systematically incomplete and uncertain.

Methods

Model description

The study uses an idealized double gyre model reproducing the North Atlantic dynamics, described in Poupon et al. (2025)[39]. This model couples an ocean physical model (Modular Ocean Model version 6)[62], developed by the Geophysical Fluid Dynamics

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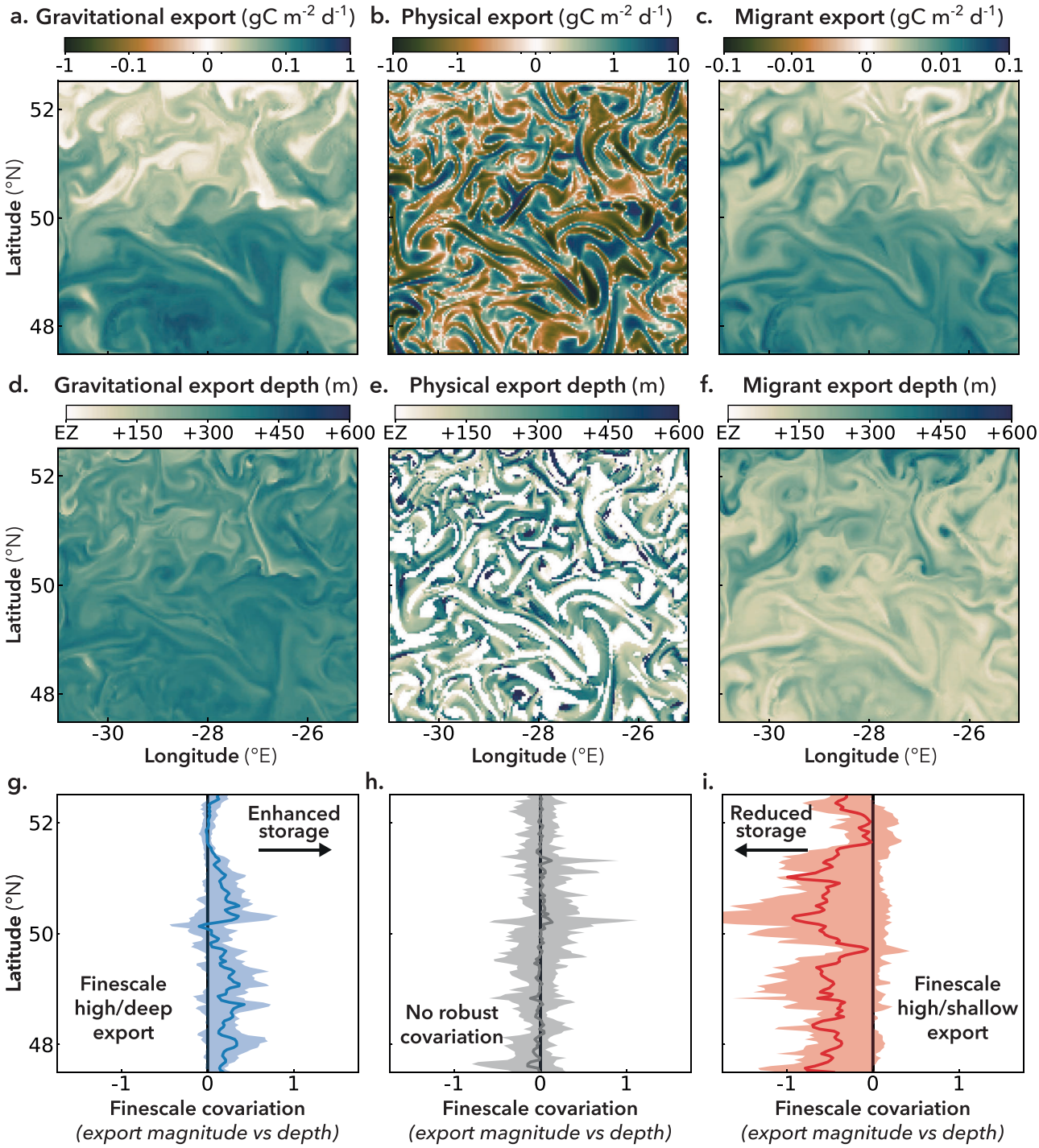


Fig. 5 | Finescale carbon export and export depth covariation. Snapshots of carbon export by (**a**) gravitational, **b** physical, and **c** migrant pumps on 10 March in the subpolar biome (31°W–25°W, 48.5°N–52.5°N; same region as Fig. 2a, b). Positive values indicate export of carbon from the euphotic to the mesopelagic zone, whereas negative values indicate a supply of carbon from the mesopelagic into the euphotic zone. Snapshots of the corresponding carbon export depth for (**d**) gravitational, **e** physical, and **f** migrant pumps, respectively, with larger values indicating deeper export relative to the base of the euphotic zone (EZ). **g**–**i** Meridional profiles of finescale covariation between carbon export and export depth (see Methods), where negative values correspond to finescale structures that jointly increase export magnitude and decrease export depth (reducing carbon storage), and positive values indicate finescale patterns in which high export is associated with deep export depths (enhancing storage).

Laboratory, with a biogeochemical module detailed below. The model domain is a box 3570 km side length and 4 km deep, centered around 40°N. To better represent fine-scale dynamics, we increased the horizontal resolution from 9 km to 3 km compared to Poupon et al. (2025)[39], with updated model parameters available in Supplementary Table S1. This 3 km model is initialized from outputs of the 9 km model run for 100 years. The 3 km model is then spun-up for 25 years. The surface of the model is continuously forced by an idealized seasonal cycle, including a sinusoidal zonal wind profile inducing a double gyre circulation.

The physical model is coupled to the COBALTv2-DVM (Carbon, Ocean Biogeochemistry and Lower Trophics version 2 with Diel

Vertical Migration) biogeochemical module[39,63]. This module features 45 biogeochemical tracers, including the main nutrients (i.e., nitrogen, phosphorus, silica, and iron) and a planktonic ecosystem composed of 3 phytoplankton and 5 zooplankton, 3 non-migrating and 2 migrating. This module represents the dynamics of suspended particulate organic carbon, in the form of a planktonic ecosystem, sinking organic carbon in the form of detritus produced by phytoplankton aggregation or zooplankton egestion and mortality. The sinking speed of particles is 100 m d^{-1} [64]. Their remineralization rate ranges between 0.02 and 0.5 d^{-1} as a function of temperature and oxygen concentration[65]. In addition, particle composition modulates remineralization: the fractions of calcium carbonate and biogenic silica associated with organic particles reduce remineralization rates, reproducing a ballast/shielding effect that enhances particle transfer to depth[63]. Mineral-poor particles remineralize shallowly, whereas mineral-rich particles sink deeper, producing a spectrum of effective particle labilities and export depths. The module also represents dissolved inorganic carbon and four pools of dissolved organic carbon (labile, semi-labile, refractory, and background ultrarefractory). Labile dissolved organic carbon is remineralized over a few days, semi-labile over 90 days, and refractory over 10 years. The background DOC pool, with a prescribed concentration of 46 μmol kg^{-1}, represents the refractory to ultrarefractory fraction of Atlantic DOC, non-reactive over the timescales considered in our experiment, as its degradation times exceed those of the overturning circulation. The COBALTv2-DVM module also explicitly represents the diel vertical migration of zooplankton as well as their compartmentalized physiology that temporally and spatially decouples ingestion, digestion, respiration, and growth.

Model biomes

The model replicates the dynamics of a low-productivity, low-seasonality subtropical biome and a high-productivity, high-seasonality subpolar biome typical of the North Atlantic Ocean. The subtropical biome is defined by an annual mean surface chlorophyll concentration below 0.15 mg m^{-3}. The subpolar biome is defined by a mean chlorophyll concentration above 0.35 mg m^{-3} and a mean mixed layer depth below 200 meters, to exclude model regions where deep convection is too strong and representative of dynamics occurring in the Norwegian and Greenland seas (see Poupon et al. 2025[39]).

Observations and validation

To evaluate the spatial and temporal variability of the biogeochemical dynamics simulated by the double-gyre model, we use satellite, ship-based, and float observations. The mixed layer depth is calculated from hydrographic profiles collected between 1941 and 2022[66], migrating zooplankton biomass from LIDAR measurements between 2007 and 2019[67], and nets collected at the Bermuda Atlantic Time Series (BATS) station between 1994 and 2010[68]. Net primary productivity estimates are produced by three algorithms (Eppley, CbPM, and CAFE) applied to Moderate-Resolution Imaging Spectroradiometer measurements between 2002 and 2023[69–71], and validated against in-situ measurements by the original authors. POC profiles come from optical backscatter measurements collected by Biogeochemical-Argo floats between 2010 and 2021[72], DIC profiles from the GLODAPv2.2023 database[73], and DOC profiles from a GEOTRACES compilation of ship measurements ranging from 1994 to 2021[74].

Framework description

Carbon remineralization and storage. In our model, total organic carbon remineralization (gC m^{-3} yr^{-1}), which produces dissolved inorganic carbon, is the sum of particulate and dissolved organic carbon remineralization by bacteria and the respiration of migrating and non-migrating plankton:

$$\begin{aligned}\text{Remineralization}(z) = {} & \text{POC remineralization}(z) + \text{DOC remineralization}(z) \\ & + \text{Biological respiration}(z)\end{aligned} \tag{1}$$

We derive carbon storage (gC m^{-3}) as the product of this remineralization rate and carbon storage time (yr) in subtropical and subpolar biomes from the Ocean Circulation Inverse Model (OCIM)[6]:

$$\text{Storage}(z) = \text{Remineralization}(z) \times \text{Storage time}(z) \tag{2}$$

The amount of carbon stored in the water column corresponds to the vertical integral of this storage profile. Because the idealized geometry of our model does not allow a one-to-one spatial mapping to OCIM, we approximate biome-scale storage by multiplying the biome-mean storage time profile from OCIM with the biome-mean remineralization profile from our model. Evaluating the difference between the mean of the product ($\overline{\int \text{Remineralization}(z) \times \text{Storage time}(z)\,dz}$) and the product of the means ($\overline{\int \text{Remineralization}(z)\,dz} \times \overline{\text{Storage time}(z)}$) using OCIM output indicates that this approximation leads to a modest underestimation of vertically integrated storage on the order of 3–5%.

We define carbon export as the amount of carbon remineralized below the euphotic zone depth (z_{eu}), defined as 0.1% of the photosynthetically active radiation at the surface[75]:

$$\text{Export} = \int_{z_{\text{eu}}}^{z_{\text{bot}}} \text{Remineralization}(z)\,dz \tag{3}$$

z_{bot} is the model bottom depth. We further define mean export depth (m) as the remineralization-weighted depth of exported carbon:

$$z_{\text{export}} = \frac{1}{\text{Export}} \int_{z_{\text{eu}}}^{z_{\text{bot}}} z\,\text{Remineralization}(z)\,dz \tag{4}$$

The mean remineralization depth associated with each export pathway (see following sections) is computed using the same expression, applied to the remineralization profile sustained by that pathway.

Carbon export pathways. The contribution of export pathways (gravitational, physical, migrant) to carbon remineralization is calculated by considering that their organic carbon supply balances the remineralization flux:

$$\text{Biological respiration} = \text{Migrant respiration} + \text{Non migrant respiration} \tag{5}$$

$$\begin{aligned}\text{POC remineralization} = {} & -w_g \frac{\partial \text{POC}}{\partial z} - \mathbf{v} \cdot \boldsymbol{\nabla} \text{POC} + K_H \nabla_H^2 \text{POC} + \frac{\partial}{\partial z}\left(K_z \frac{\partial \text{POC}}{\partial z}\right) \\ & + \text{POC production} - \frac{\partial \text{POC}}{\partial t}\end{aligned} \tag{6}$$

$$\begin{aligned}\text{DOC remineralization} = {} & -\mathbf{v} \cdot \boldsymbol{\nabla} \text{DOC} + K_H \nabla_H^2 \text{DOC} + \frac{\partial}{\partial z}\left(K_z \frac{\partial \text{DOC}}{\partial z}\right) \\ & + \text{DOC production} - \frac{\partial \text{DOC}}{\partial t}\end{aligned} \tag{7}$$

DOC and POC are the concentrations of dissolved and particulate organic carbon, $\mathbf{v} = (u, v, w)$ is the fluid velocity, w_g is the gravitational sink velocity relative to the fluid, and K_H and K_z are the horizontal and vertical diffusion coefficients. DOC and POC are produced by both migrating and non-migrating animals (egestion, excretion, mortality). Below the euphotic zone, production by non-migrant animals is marginal and can be neglected. On an annual average, the gyre is close to equilibrium. As a result, $\partial \text{DOC}/\partial t$ and $\partial \text{POC}/\partial t$ are also marginal and

can be neglected. Thus, Eq. (1) can be written as:

$$\text{Remineralization}(z) = \text{Gravitational pathway}(z) + \text{Physical pathway}(z) + \text{Migrant pathway}(z) \tag{8}$$

with

$$\text{Gravitational pathway} = -w_g \frac{\partial \text{POC}}{\partial z} \tag{9}$$

$$\text{Physical pathway} = -\mathbf{v}\cdot\boldsymbol{\nabla}\text{POC} + K_H\nabla_H^2\text{POC} + \frac{\partial}{\partial z}\left(K_z\frac{\partial \text{POC}}{\partial z}\right) - \mathbf{v}\cdot\boldsymbol{\nabla}\text{DOC} + K_H\nabla_H^2\text{DOC} + \frac{\partial}{\partial z}\left(K_z\frac{\partial \text{DOC}}{\partial z}\right) \tag{10}$$

$$\text{Migrant pathway} = \text{Migrant respiration} + \text{Migrant POC and DOC production} \tag{11}$$

Physical pathway decomposition. The physical export pathway encompasses transport by large-scale circulation such as Ekman transport or overturning circulation (large-scale pathway), subduction induced by finescale structures such as submesoscale fronts and mesoscale eddies (eddy-subduction pathway), and transport induced by mixed-layer deepening and restratification (mixed-layer pathway).

$$\text{Physical pathway} = \text{Large scale pathway} + \text{Eddy subduction pathway} + \text{Mixed layer pathway} \tag{12}$$

We isolate the small- and large-scale advective components using a Reynolds decomposition[76]:

$$\overline{C\mathbf{v}} = \overline{C}\,\overline{\mathbf{v}} + \overline{C'\mathbf{v}'} \tag{13}$$

where $\overline{(\cdot)}$ is a Gaussian spatial mean operator with a radius of 100 km and $(\cdot)'$ denotes the small-scale residual. We isolate vertical mixing in the mixed layer from mixing in the ocean interior using an indicator function, $\text{Ind}_{z<\text{MLD}}$, which takes the value 1 for depths within the mixed layer and 0 otherwise, and vice versa for $\text{Ind}_{z>\text{MLD}}$. The three pathways of the physical pump are then:

$$\text{Large scale pathway} = -\boldsymbol{\nabla}\cdot\left(\overline{\mathbf{v}}\,\overline{\text{POC}} + \overline{\mathbf{v}}\,\overline{\text{DOC}}\right) + K_H\nabla_H^2(\text{POC}+\text{DOC}) + \frac{\partial}{\partial z}\left(K_z\left(\frac{\partial \text{POC}}{\partial z} + \frac{\partial \text{DOC}}{\partial z}\right)\right)\text{Ind}_{z>\text{MLD}} \tag{14}$$

$$\text{Eddy subduction pathway} = -\boldsymbol{\nabla}\cdot\left(\overline{\mathbf{v}'\text{POC}'} + \overline{\mathbf{v}'\text{DOC}'}\right) \tag{15}$$

$$\text{Mixed layer pathway} = \frac{\partial}{\partial z}\left(K_z\left(\frac{\partial \text{POC}}{\partial z} + \frac{\partial \text{DOC}}{\partial z}\right)\right)\text{Ind}_{z<\text{MLD}} \tag{16}$$

This framework makes it possible to quantify not only the various components of the biological carbon pump, but also their respective spatial and temporal variability across all space and time scales.

Finescale covariation factor

To highlight how finescale dynamics jointly modulate carbon export magnitude and export depth, we decomposed each variable into a large-scale background field and a finescale anomaly, using the same Gaussian spatial mean operator described above, $\overline{(\cdot)}$. For each export pathway, the export flux e and export depth d were written as $e = \overline{e} + e'$ and $d = \overline{d} + d'$, where $\overline{e}$ and $\overline{d}$ denote the large-scale components obtained via Reynolds decomposition, and e' and d' represent the finescale residuals. We then defined a finescale covariation factor that measures the local co-fluctuation of anomalies in export magnitude and export depth:

$$C = \frac{e'}{\sigma_e} \times \frac{d'}{\sigma_d} \tag{17}$$

where σ_e and σ_d are the standard deviations of the finescale export and depth anomalies over the Gaussian filter domain. Positive values of C indicate that finescale structures simultaneously enhance export magnitude and deepen export (or reduce export magnitude and shoal export), whereas negative values indicate opposite-signed anomalies.

Data availability

All the data generated in this study have been deposited on Zenodo and are available at https://doi.org/10.5281/zenodo.19482976.

Code availability

The COBALTv2-DVM model and idealized double gyre configuration used in this paper are available on GitHub at https://github.com/mpoupon/MOM6_Double_Gyreand Zenodo at https://doi.org/10.5281/zenodo.15215671.

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Author contributions

Conceptualization: MP, LR, JL. Data curation: MP. Formal analysis: MP. Funding acquisition: LR, JL. Investigation: MP. Methodology: MP. Software: MP. Supervision: LR, JL. Visualization: MP. Writing - original draft: MP. Writing - review and editing: MP, LR, JL.

Competing interests

The authors declare no competing interests.

Additional information

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